AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 7 · STUDENT LESSON

Parameters and architecture

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Declare the architecture. Count its slots. Test quality separately.

Count dense weights and biases while distinguishing model values from other quantities.

Keep one habit: Named thirteen slots → adjacent-layer width change → picture count → convention and sorting probes → independent architecture.

MAIN LESSON · SECTION 1

Which values belong to the model,and which are computed?

> Count connections and receiving-unit biases under a declared architecture.

Learning goals

I can distinguish architectural choices from **parameter** values.

I can count weights and biases in a small dense network.

I can explain why **parameter** count alone cannot establish quality.

Retrieve the prior computation

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- In the Week 6 network, which quantities change when x changes while the model stays fixed?

> **My starting explanation**

End goal

By the end, I can explain **parameter**s and **architecture** using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Distinguish architectural choices from **parameter** values

Count weights and biases in a small dense network

Explain why **parameter** count alone cannot establish quality

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Read the architecture before counting its values

> This dense2→3→1network has one independent bias per receiving unit.

Declare widths,connections and activations

Consider a dense network with 2 **input**s, 3 hidden units using ReLU, and 1 **output** unit using identity. We call its widths 2 to 3 to 1. These widths, connectivity and activation choices describe its **architecture**.

The weights and biases supply the numerical values used by that **architecture**.

A fully connected hidden unit receiving 2 **input**s needs 2 weights plus 1 bias. Three such units need 6 weights and 3 biases. The **output** receives 3 activations and needs 3 weights plus 1 bias. Total: 13 **parameter**s.

Input values and computed activations are not additional **parameter**s.

Complete layer-by-layer count

Layer	Weights	Biases	Total
2 inputs to 3 hidden units	2 times 3 = 6	3	9
3 hidden values to 1 output	3 times 1 = 3	1	4

Layer	Weights	Biases	Total
Whole network	9	4	13

State no shared weights and the bias convention

This counting rule assumes one separate bias per **output** unit and no shared or tied weights. Other **architecture**s can use different rules.

Name supplied and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Count one slot for each weight and each receiving bias

> Compute each layer separately before adding.

General dense-layer formula

For a **dense layer** with n **input** components and m **output** units, each **output** has n weights and one bias. The count is n times m plus m, or (n + 1) times m. Count each layer separately before adding.

All thirteen named parameter slots

Parameter slots; one value per named slot	Count
Hidden unit 1:w11, w12, b1	3
Hidden unit 2:w21, w22, b2	3
Hidden unit 3:w31, w32, b3	3
Output:v1, v2, v3, c	4
Total; inputs and hidden activations excluded	13

Complete width3versus4comparison

Same2-input/1-output dense convention · Hidden weights · Hidden biases · Output weights · Output bias · Total

> **Same2-input/1-output dense convention:** Hidden width 3
>
> **Hidden weights:** 6
>
> **Hidden biases:** 3
>
> **Output weights:** 3
>
> **Output bias:** 1
>
> **Total:** 13

> **Same2-input/1-output dense convention:** Hidden width 4
>
> **Hidden weights:** 8
>
> **Hidden biases:** 4
>
> **Output weights:** 4
>
> **Output bias:** 1

Total: 17

Same2-input/1-output dense convention: What one added hidden unit changes

Hidden weights: +2

Hidden biases: +1

Output weights: +1

Output bias: +0

Total: +4

Runnable counting aid and assumptions

```
def dense_count(inputs, outputs):
    return inputs * outputs + outputs

print(dense_count(2, 3))   # 9
print(dense_count(3, 1))   # 4
print(dense_count(2, 3) + dense_count(3, 1))  # 13
```

The helper assumes positive integer widths and the bias convention stated above. It is a counting aid, not a model trainer. A ReLU has no fitted numerical **parameter**s in this example.

Connect each visual connection to a counted value

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

Separate a parameter value from an architectural change

> A training setting is a third kind of choice.

Current quantity-sorting task

- Classify hidden activation 0.7, weight -2, hidden width 3 and learning rate 0.1 as computed value, **parameter**, **architecture** choice or **training** setting.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured added hidden unit

- Change: Increase hidden width from 3 to4.

▼ **Check after predicting**

Prediction: First layer 12 **parameter**s; **output** layer 5; total 17.

Reason: Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

See why one added unit changes two layers

> Count its incoming connections,bias and outgoing connection.

Complete pictured architecture and counting contract

Count trainable weights and biases separately from activation functions.

Dense toy **architecture** 2 **input**s→3 hidden units→1 **output**. Every hidden and **output** unit has one bias; hidden activation is ReLU.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture and main count use the same fully connected 2→3→1 architecture:6 hidden weights+3 hidden biases+3 output weights+1 output bias=13. ReLU contributes no fitted parameter here. Increasing only hidden width 3→4 adds2 incoming weights, 1 hidden bias and 1 outgoing weight:4 more, total 17. It changes both adjacent layers; the output bias remains one.

Parameter values, computed activations and learning rate are not additional architectural connection slots.

Trace both layer counts

AI ACADEMY / AI-5/7 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Count first layer	2×3=6 weights plus 3 biases gives 9 parameters.
2. Count output layer	3×1=3 weights plus 1 bias gives 4 parameters.
3. Total	Total trainable parameters=9+4=13; ReLU adds no parameters here.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Count one weight for each **input**-to-unit connection and one bias for each receiving unit. Two **input**s connecting to three hidden units make six weights.

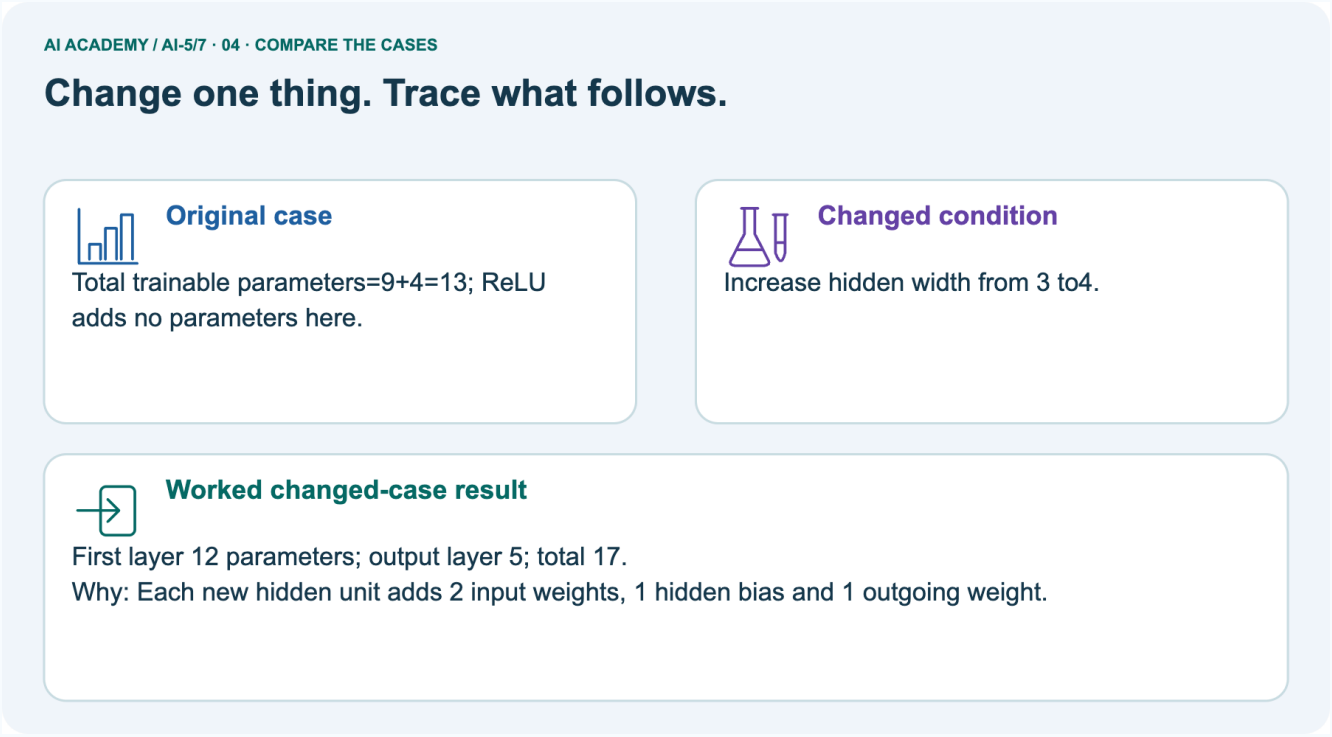
The three hidden **output**s connect to one **output** unit through three more weights. ReLU applies a fixed function and adds no adjustable **parameter** here.

Architecture determines the connection pattern; learned values occupy those connections.

Predict hidden width4

- Change one thing: Increase hidden width from 3 to4. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the increment and quality limit

- **Parameter** count describes capacity and storage structure, not accuracy, intelligence or guaranteed generalisation.

- Read the labelled picture: How many new **parameter**s did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try the complete wider-network count

Decide what a change alters

PROPOSED CHANGE	TYPE OF CHANGE	CONSEQUENCE
Change one weight 2 to 3	Parameter value	Same architecture
Use 4 hidden units	Architecture width	Different parameter count
Use learning rate 0.1	Training setting	No extra model weight

Each row is a separate proposed change. A parameter value, an architectural choice and a training setting answer different questions about the system.

For a dense 2 to 4 to 1 network with one bias per unit, count each layer and the total. How many more parameters does it have than 2 to 3 to 1?

▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Parameters and architecture

The facts we will use

AI ACADEMY / AI-5/7

Trainable parameter count

	weights	biases	total
input to hidden	6	3	9
hidden to output	3	1	4

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/7

Parameters and architecture

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count first layer

2 x 3=6 weights plus 3 biases gives 9 parameters.



2. Count output layer

3 x 1=3 weights plus 1 bias gives 4 parameters.



3. Total

Total trainable parameters=9+4=13; ReLU adds no parameters here.

CHANGE ONE THING

Increase hidden width from 3 to4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Repair “more parameters means better”and unstated conventions

> A count describes a model structure,not its measured performance.

Same architecture,different values and fair width comparisons

Two networks can share an **architecture** and still make different **prediction**s because their **parameter** values differ. Conversely, two **architecture**s can happen to produce identical **output**s on the few cases tested.

A count describes the model, not how well it has learned.

When comparing widths, keep the task, feature schema, **training** data, validation cases and evaluation rule fixed. Record the **training** budget and seed as well.

More **parameter**s can increase flexibility, but they can also increase resource use and over**fitting** risk. Neither outcome follows from counting alone.

Recount with every bias omitted

- Recount the 2 to 3 to 1 model if every bias is omitted. State what the original count assumed.

> **State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

Practise counting and sorting with less help

> State the convention used for each answer.

Current task and response

- A **dense layer** has 4 **input**s and 2 **output**s with one bias per **output**. Count weights, biases and total.

> **For a dense 2 to 4 to 1 network with one bias per unit, count each layer and the total. How many more parameters does it have than 2 to 3 to 1?**

> **Explain your prediction to a partner before comparing results.**

Supported W2record

Use workbook W2 for the connected example "**Parameter**s and **architecture**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for **Parameter**s and **architecture**. Proposed explanation: “Count one bias per **input**”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured parameter-increment reasoning

- How many new **parameter**s did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Reserve the independent new architecture

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Sort the quantities

Check your reasoning

Explain the picture

Run a complete parameter-slot and quantity-sorting audit

> Every probe states its architecture convention or missing information.

Current sorting task and runnable counting aid

Classify hidden activation 0.7, weight -2, hidden width 3 and learning rate 0.1 as computed value, **parameter**, **architecture** choice or **training** setting.

```
def dense_count(inputs, outputs):
    return inputs * outputs + outputs

print(dense_count(2, 3))   # 9
print(dense_count(3, 1))   # 4
print(dense_count(2, 3) + dense_count(3, 1))  # 13
```

Complete count,boundary,sorting and missing-definition probes

Complete counting/sorting probe	Expected answer and limit
Normal dense 2→3→1 with one bias per unit	9+4=13
Smallest positive layer width 1→1	1 weight+1 bias=2
Same2→3→1 but omit all biases	6+3=9; changed convention, not the original count
Activation value.7	Computed intermediate; not a separately fitted parameter
Weight−2	Parameter value; changing it preserves the architecture
Hidden width 3	Architecture choice; also a model-design hyperparameter
Learning rate.1	Training setting; not another model weight
Missing bias/shared-weight convention	Request the architecture definition; do not assume this dense formula always applies

Complete counting/sorting probe	Expected answer and limit
Invalid layer width 0or–1	Outside positive-integer-width contract; the simple helper does not validate this
Two models both have 13 parameters	Count alone does not prove same architecture, values, predictions or quality

Count,sort,check and bound the conclusion

1. Count named slots on paper first, then use dense_count on valid positive integer widths to check each layer. Sort all supplied quantities and justify their roles. Recount the no-bias variant explicitly. This helper only computes a formula; it does not validate widths, train a network or measure speed, memory or accuracy.

   A fair quality comparison needs fixed task, data/splits/evaluation and recorded training budget/settings; more parameters alone settle none of those results.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: count a different dense architecture

> Recount both adjacent layers after the supplied hidden-width change.

Complete new independent architecture and convention

A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the **output** layer, with one bias per unit. Give each layer count and the total.

If hidden width becomes 4, recount and explain why activation values are excluded.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Count a new architecture independently

Attempt this without hints

A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the **output** layer, with one bias per unit. Give each layer count and the total.

If hidden width becomes 4, recount and explain why activation values are excluded.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save an architecture definition alongside parameter values

> Someone else should know exactly which computation the count describes.

Project artifact and evidence table

Add a labelled **parameter**s and **architecture evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish architectural choices from **parameter** values.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Keep actual evidence and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one counted slot and one unmeasured claim

> Explain why a larger count cannot replace an evaluation.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Distinguish architectural choices from **parameter** values

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Parameter	A numerical model value that a fitting procedure may adjust
Architecture	The arrangement of layers, connections and activation choices
Dense layer	A layer in which each output unit has a weight for every input component
Hyperparameter	A setting chosen for a model or training procedure rather than fitted as an ordinary weight

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- A **dense layer** has 4 **input**s and 2 **output**s with one bias per **output**. Count weights, biases and total.
- Recount the 2 to 3 to 1 model if every bias is omitted. State what the original count assumed.
- A team says its 26-**parameter** model must outperform a 14-**parameter** model. Give a fair check and one reason the claim is unsupported.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent3→2→2architecture and wider variant for the first individual attempt.

Current explanation and response

- A reproducible description needs ordered **input** features, layer widths, connections, activations and the exact **parameter** values or a way to recover them.

  A description of **training** also needs the data split and **training** settings. The label small neural network does not specify a computation.

- Next week defines a loss so that we can measure how a particular set of **parameter**s performs on labelled examples.

- A team says its 26-**parameter** model must outperform a 14-**parameter** model. Give a fair check and one reason the claim is unsupported.

> **Two teams both report 13 parameters. What would you request before claiming they built the same model?**

> **My limited claim and a useful next check**

▼ Extra practice · Revisit the complete trace

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Decide what a change alters

PROPOSED CHANGE	TYPE OF CHANGE	CONSEQUENCE
Change one weight 2 to 3	Parameter value	Same architecture
Use 4 hidden units	Architecture width	Different parameter count
Use learning rate 0.1	Training setting	No extra model weight

Each row is a separate proposed change. A parameter value, an architectural choice and a training setting answer different questions about the system.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Count trainable weights and biases separately from activation functions.

By the end, I can	How I will show it
Distinguish architectural choices from parameter values	Explain the deciding step, show a trace or test, and state one limit.
Count weights and biases in a small dense network	Explain the deciding step, show a trace or test, and state one limit.
Explain why parameter count alone cannot establish quality	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Count trainable weights and biases separately from activation functions.
- Which labelled step matches this detail? Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here.
- 1. Total
- 2. Count **output** layer
- 3. Count first layer
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Total Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here. Count one weight for each **input**-to-unit connection and one bias for each receiving unit.

Two **input**s connecting to three hidden units make six weights. The three hidden **output**s connect to one **output** unit through three more weights. ReLU applies a fixed function and adds no adjustable **parameter** here.

Architecture determines the connection pattern; learned values occupy those connections.

Changed-case question and optional hint

- Changed-case practice: Increase hidden width from 3 to4. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. First layer 12 **parameter**s; **output** layer 5; total 17.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: First layer 12 **parameter**s; **output** layer 5; total 17. Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

Repair and retain the record

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Activation functions (ai-5/4). Goal: Trace identity ReLU and sigmoid on a numerical score. Short practice: Which original **input**s cannot be distinguished by this network's final **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Representing **input**s as vectors (ai-5/1). Goal: Encode a record using a fixed feature order and units. Short practice: Why does equal total brightness not make A and B identical **input**s?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

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<summary>Reference · Six-picture concept journey</summary>

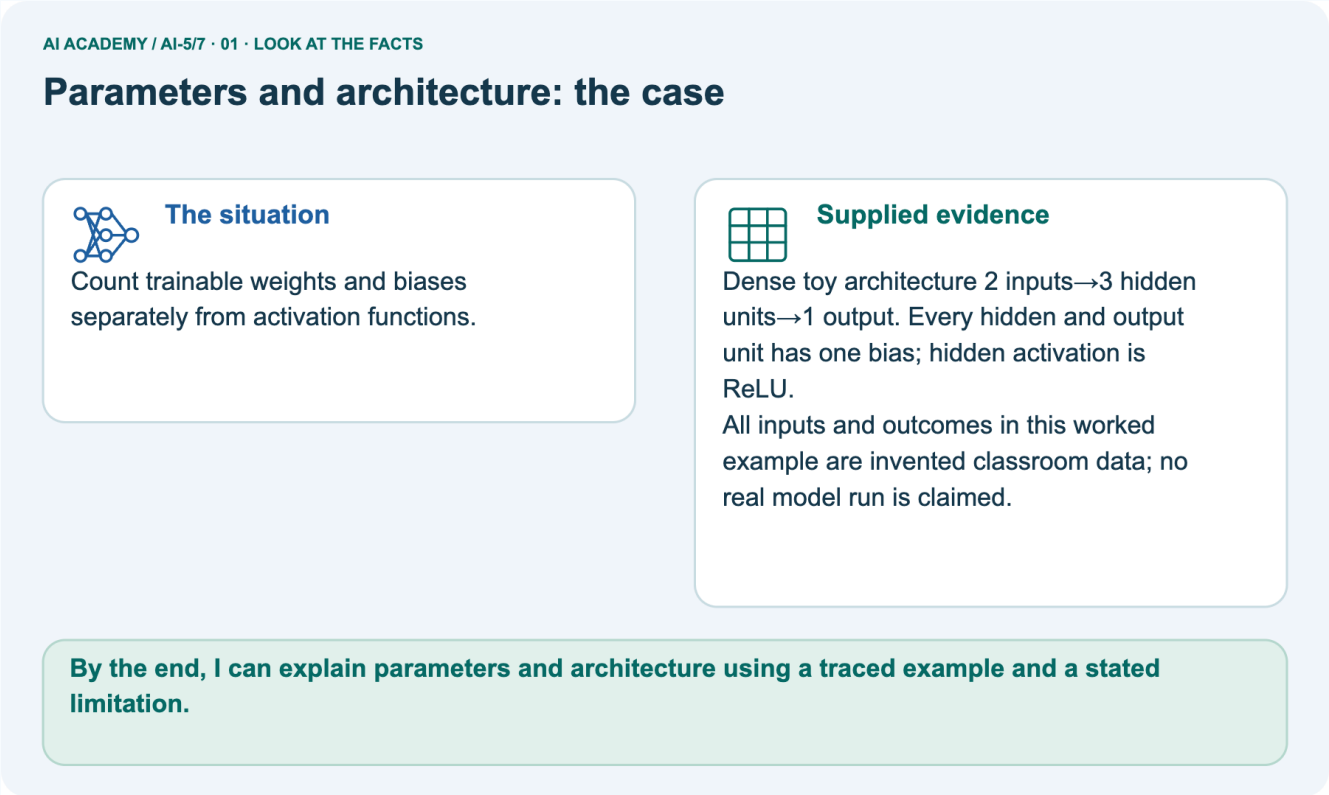
</details>

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Parameters and architecture: the case



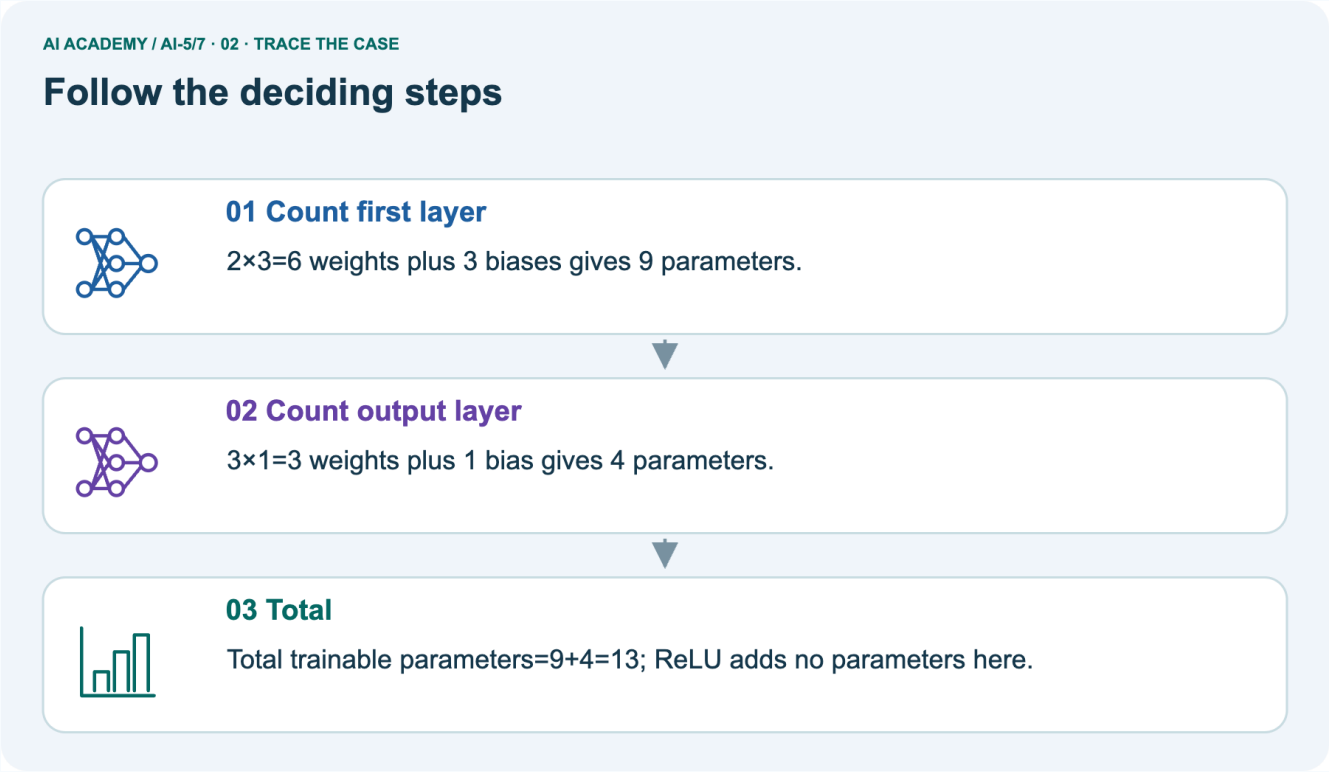
Follow the labelled picture with the matching explanation. Open the vector diagram

Count trainable weights and biases separately from activation functions. Supplied facts: Dense toy **architecture** 2 **input**s→3 hidden units→1 **output**. Every hidden and **output** unit has one bias; hidden activation is ReLU.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

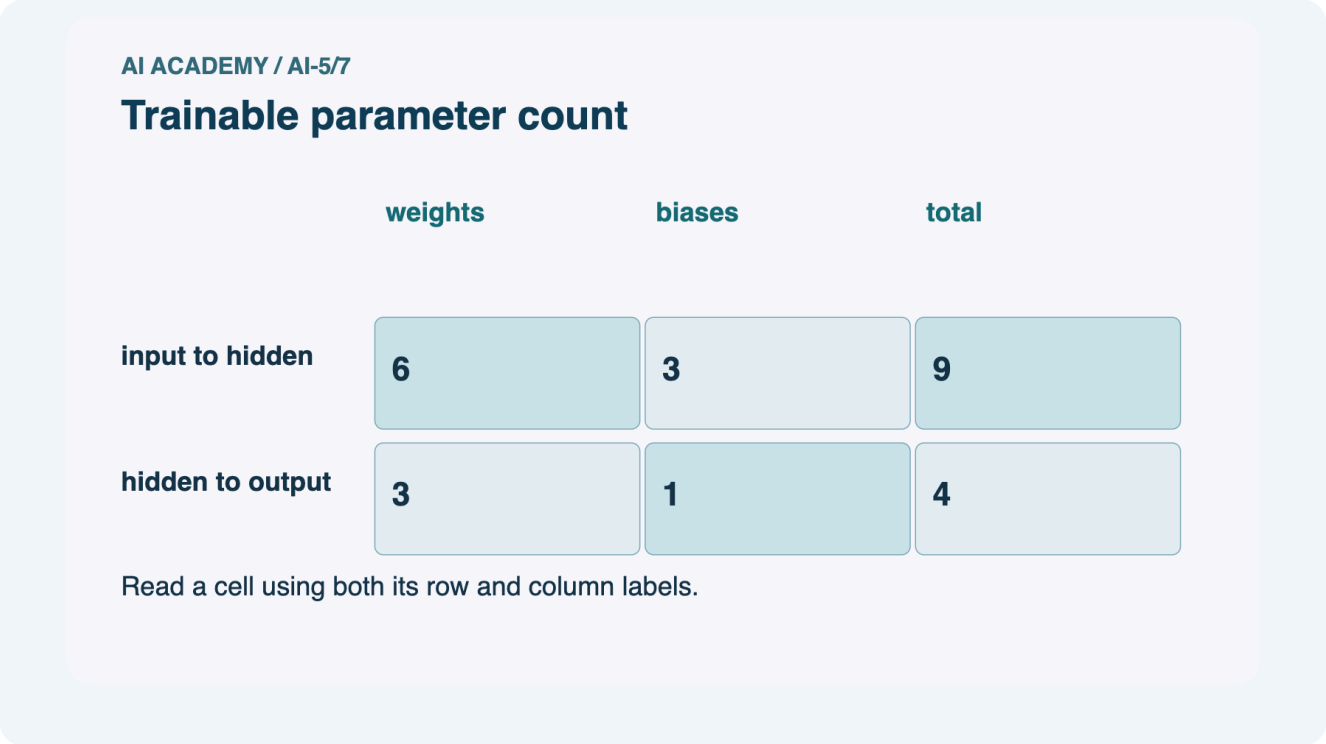


Follow the labelled picture with the matching explanation. Open the vector diagram

Count first layer: 2×3=6 weights plus 3 biases gives 9 **parameter**s. → Count **output** layer: 3×1=3 weights plus 1 bias gives 4 **parameter**s. → Total: Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here.

Point to each numbered stage and explain which recorded **input** it uses.

03. Trainable parameter count

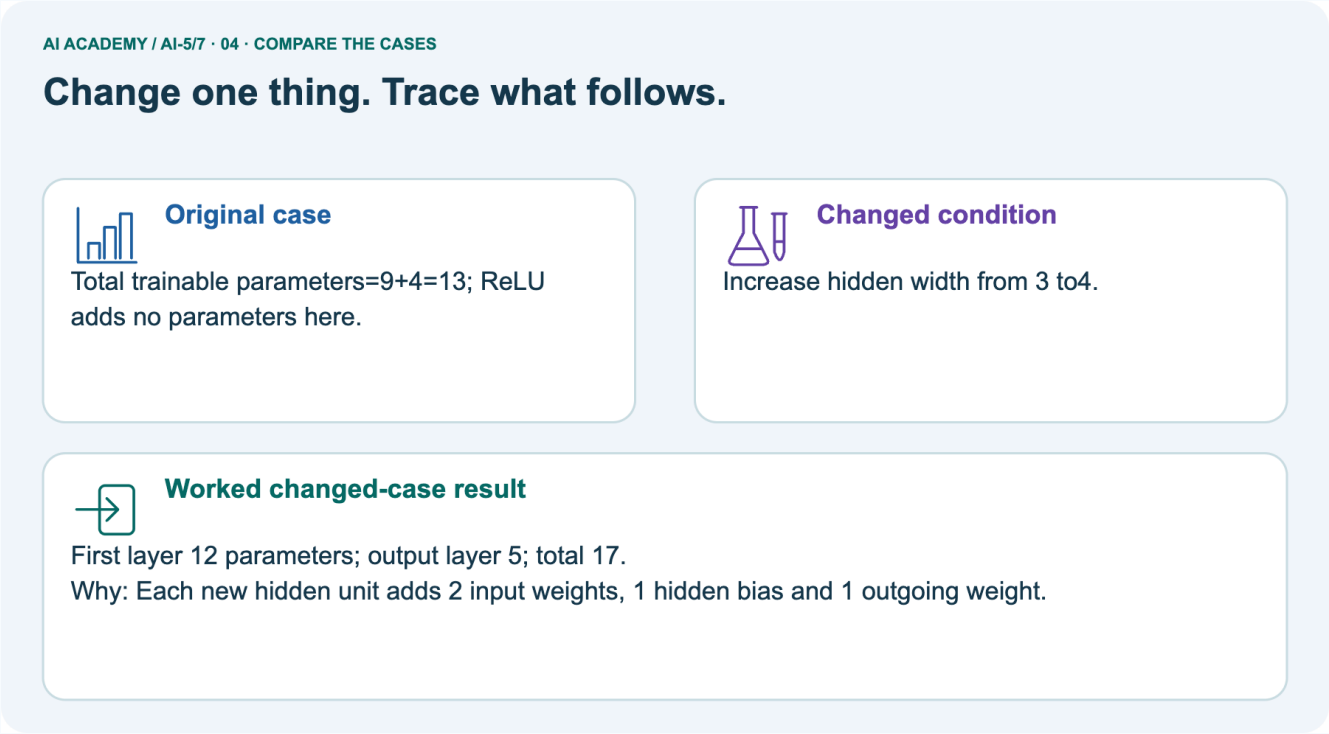


Follow the labelled picture with the matching explanation. Open the vector diagram

Trainable **parameter** count. **input** to hidden: weights = 6, biases = 3, total = 9. hidden to **output**: weights = 3, biases = 1, total = 4.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



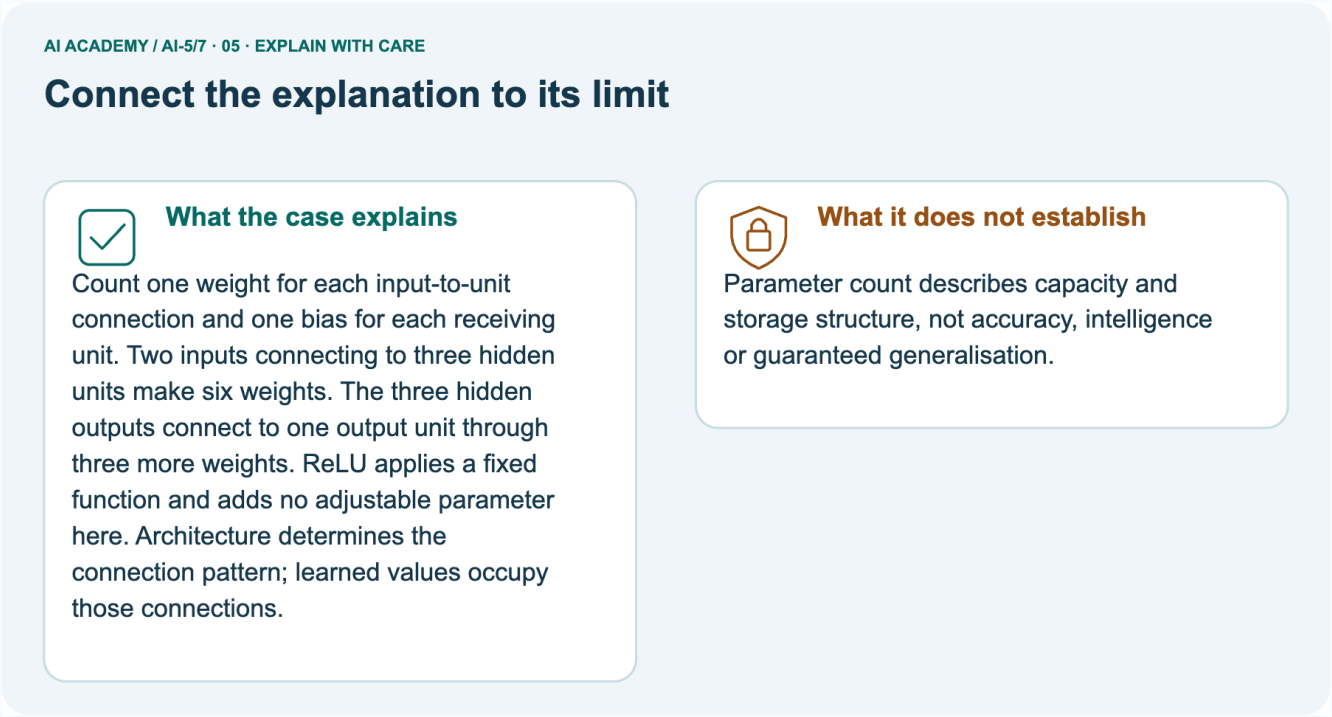
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here. Changed condition: Increase hidden width from 3 to4. Expected result: First layer 12 **parameter**s; **output** layer 5; total 17.

Reason: Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

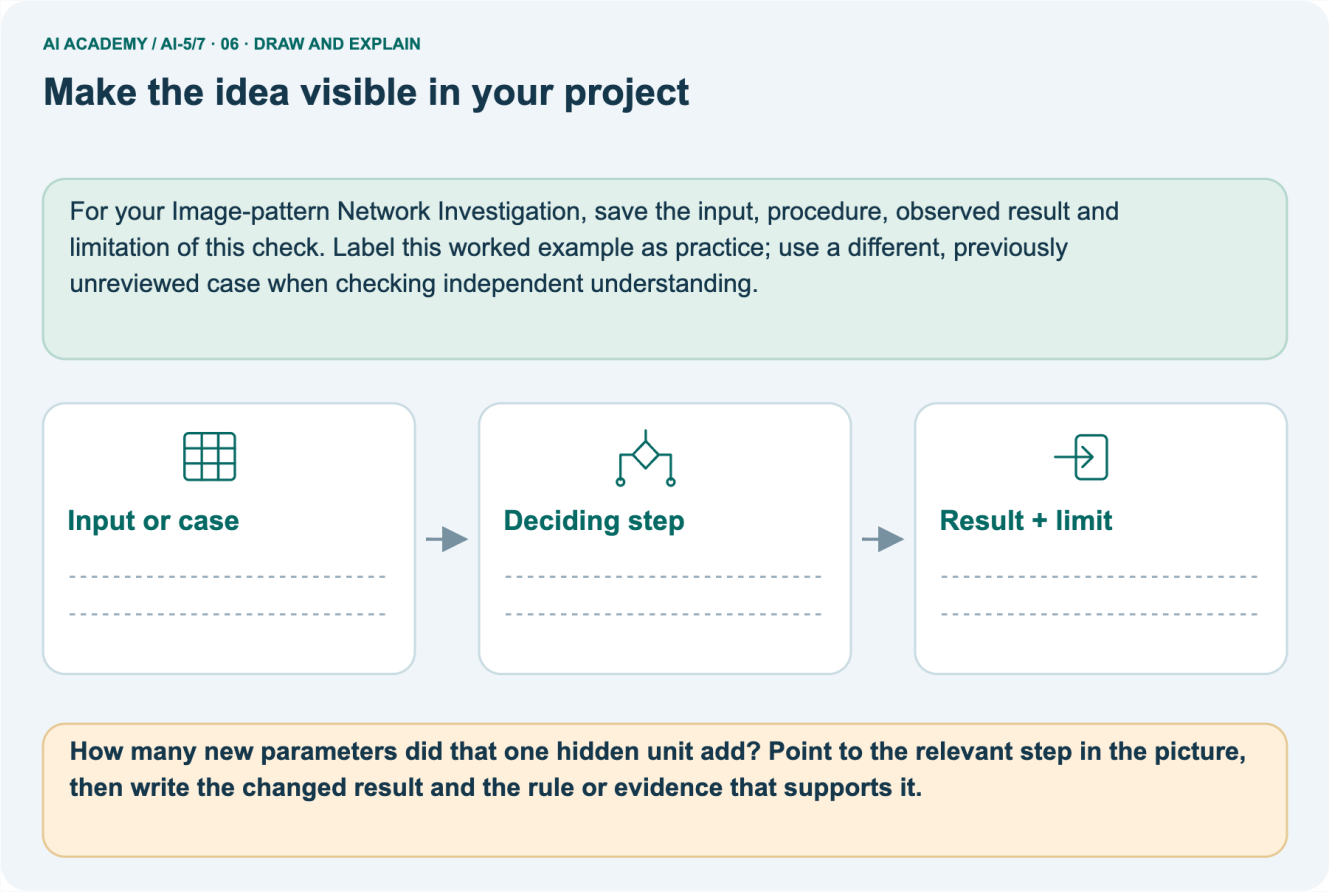
Count one weight for each **input**-to-unit connection and one bias for each receiving unit. Two **input**s connecting to three hidden units make six weights.

The three hidden **output**s connect to one **output** unit through three more weights. ReLU applies a fixed function and adds no adjustable **parameter** here.

Architecture determines the connection pattern; learned values occupy those connections. Limit: **Parameter** count describes capacity and storage structure, not accuracy, intelligence or guaranteed generalisation.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many new **parameter**s did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Parameters and architecture: the case

02. Follow the deciding steps

03. Trainable parameter count

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.